

Project Proposal & System Design Methodology

Hybrid RAG & Fine-Tuning for Enterprise Customer Support Systems

Project Phase: Milestone 1 Technical Specification
(Tasks 1.1 & 1.4)

Target Domain: E-Commerce Enterprise Customer Service

Base Foundation Model: Qwen/Qwen2.5-1.5B-Instruct

Document Version: 1.0 (Formal Capstone Deliverable)

1. Executive Overview & Problem Formulation

Modern automated customer support systems require both strict fidelity to authoritative corporate policies and conversational adaptability across varied customer inquiries. Current enterprise solutions suffer from two core failure modes: **(1) Parametric Hallucination:** Unassisted LLMs invent business policies, erroneous return windows, or fictitious deadlines; and **(2) Retrieval Dilution in Naive RAG:** Dense vector similarity search directly over raw customer inputs often retrieves irrelevant policy clauses because user queries are rife with emotional noise, rambling context, or multiple intents. This project proposes and implements a **Hybrid RAG architecture** that decouples query intent routing from factual synthesis, combining Parameter-Efficient Fine-Tuning (PEFT) LoRA with structured SOP vector retrieval.

2. Task 1.1: Define Project Scope & Objectives

2.1 Business & Technical Objectives

- **Eliminate Policy Hallucination:** Reduce factual hallucination rates from >30% down to <5% by grounding all responses in verified corporate Standard Operating Procedures (SOPs).
- **High Precision Intent Routing:** Fine-tune a specialized router to output valid JSON representations with >98% format adherence and robust intent categorization.
- **Targeted Document Retrieval:** Overcome naive retrieval dilution by querying the vector database with canonical policy headers, raising retrieval precision to >80%.
- **Computational Efficiency:** Deliver an architecture operable under CPU and low-resource environments using parameter-efficient LoRA adapters and bfloat16 precision.

2.2 Key Performance Indicators (KPIs) & Target Metrics

Metric / KPI	Baseline Target	Naive RAG Target	Hybrid RAG (Goal)
Format Adherence (%)	≥ 95.0%	≥ 95.0%	100.0% (Strict JSON)
Retrieval Precision (%)	0.0% (N/A)	~60.0%	≥ 80.0%
ROUGE-L (F1)	~0.18	~0.18	≥ 0.21 (+15%)
BLEU Score	~0.014	~0.018	≥ 0.040 (+120%)
Hallucination Rate (%)	> 30.0%	~15.0%	< 5.0%
Adversarial Robustness	Untested	Poor (<50%)	≥ 90.0%

2.3 Project Scope Boundaries

In Scope: Customer support intents covering 27 distinct operations across 11 high-level categories (orders, shipping, returns, cancellations, refunds, accounts, payment methods, billing, technical support, privacy, and operating hours); a corpus of 13 official corporate policy markdown documents; parameter-efficient LoRA adapter training; dense vector indexing; and comparative evaluation on clean held-out splits and adversarial query subsets.

Out of Scope: External live database API transactional executions (e.g. charging credit cards), continuous online streaming retraining, and multi-modal voice processing.

3. Data Understanding & Preparation Strategy

Dataset Acquisition & Structure: The system utilizes the bitext/Bitext-customer-support-llm-chatbot-training-dataset comprising 26,872 customer interactions labeled with instruction, intent, and category. To establish a balanced, high-signal training distribution, stratified sampling selects **3,902 representative examples** guaranteeing proportional representation across all 27 intents.

Corporate Policy Corpus: Standard Operating Procedures (SOPs) are curated across 13 markdown documents containing exact business rules: refund_policy.md, order_tracking.md, shipping_delays.md, product_return.md, cancel_order.md, subscription_cancellation.md, payment_methods.md, billing_disputes.md, account_recovery.md, password_reset.md, technical_troubleshooting.md, data_privacy.md, and working_hours.md.

Data Splitting Protocol: To prevent data leakage, the sampled dataset is partitioned using strict stratified splitting: **70% Train** (2,731 samples), **20% Validation** (780 samples), and **10% Held-Out Test** (391 samples). The test split is completely quarantined and utilized exclusively for final zero-leakage evaluation.

4. Task 1.4: System Design Methodology

The technical methodology adopts a comparative experimental lifecycle consisting of three progressive architectural milestones:

4.1 Tri-Architecture Comparative Lifecycle

Pipeline Phase	Architecture	Operational Mechanism	Evaluation Objective
Phase 1	Baseline Model (Zero-Shot)	Direct prompt inference on Qwen2.5-1.5B-Instruct without retrieval.	Quantify baseline hallucination rate and lack of policy grounding.
Phase 2	Solution V1 (Naive RAG)	Dense similarity search (MiniLM) using raw customer query → Top-k SOP context → LLM.	Measure retrieval precision and isolate retrieval dilution failure modes.
Phase 3	Solution V2 (Hybrid RAG)	LoRA fine-tuned intent router → Canonical SOP search → Grounded generation.	Validate intent accuracy, retrieval precision gains, and hallucination reduction.

4.2 Fine-Tuning Methodology (PEFT LoRA)

To adapt the base LLM into a deterministic intent classification router without the memory overhead of full parameter updates, **Low-Rank Adaptation (LoRA)** is applied to the key projection modules:

- **Target Modules:** Query and Value attention projection layers (q_proj, v_proj).
- **Hyperparameters:** Rank $r = 16$, Alpha $\alpha = 32$, Dropout 0.05.
- **Instruction Template:** ChatML formatting wrapping user instructions into valid target JSON: {"intent": "", "category": ""}.
- **Optimization:** AdamW optimizer with learning rate $2e-4$, gradient accumulation, and evaluation validation loss checkpoints.

4.3 Retrieval-Augmented Generation Architecture

- **Embedding Engine:** sentence-transformers/all-MiniLM-L6-v2 maps text into a 384-dimensional dense semantic space.
- **Vector Database:** Persistent ChromaDB stores all 13 corporate SOP documents with metadata annotations.
- **Canonical Mapping:** The router's predicted intent is mapped directly to authoritative SOP search strings (e.g., track_order → 'Order Tracking # Track Order Status Delivery'), guaranteeing that ChromaDB retrieves the exact clause needed for answer generation.

5. Evaluation Framework & Validation Protocol

The evaluation protocol enforces zero data contamination and comprehensive metric tracking:

- **Format Adherence Rate:** Validates that the router emits strictly parsable JSON compliant with schema specifications.
- **Intent Exact & Fuzzy Match:** Measures exact categorical alignment as well as semantic token overlap between predicted and ground-truth intents.
- **Retrieval Precision:** Verifies whether the document returned by ChromaDB matches the ground-truth SOP for the inquiry.
- **Lexical & Semantic Quality:** ROUGE-1, ROUGE-L, and BLEU scores evaluate generated response fidelity against policy ground truth.
- **Adversarial Stress Testing:** Evaluates router resilience on customer inquiries containing intense sentiment, negation words ('never', 'cannot', 'broken'), and hedging expressions.

6. Risk Assessment, Governance & Mitigation

Identified Risk	Severity	Impact	Mitigation Strategy
Out-of-Domain Query	Medium	Router predicts incorrect intent; irrelevant SOP retrieved.	Confidence thresholding and fallback routing to human escalation matrix.
Memory / Latency Constraint	High	Model out-of-memory crashes on resource-limited CPU hardware.	bfloat16 precision loading, LoRA adapter fusion, and multi-threaded CPU execution.
Format Parsing Failure	Medium	Model outputs conversational preamble breaking JSON extraction.	ChatML system prompting with regex extraction fallback (re.search(r'\{.*?\}')).
Policy Hallucination	Critical	Misleading customers on refund rules, leading to brand liability.	Strict system grounding instructions prohibiting extrapolation beyond context.

7. Implementation Roadmap & Milestones

Milestone	Target Artifacts	Deliverable File
Milestone 1: Foundations	Tasks 1.1–1.4: Scope, EDA, Sampling, Methodology	Project_Proposal_and_Methodology.pdf, Data_Understanding_and_EDA.ipynb
Milestone 2: Data & Baseline	Tasks 2.1–2.4: ChatML Prep, Tokenization, Baseline Eval	Data_Preparation.ipynb, Baseline_Model_Evaluation.ipynb
Milestone 3: Naive RAG	Tasks 3.1–3.4: ChromaDB Indexing, V1 Evaluation	RAG_Implementation.ipynb, Solution_V1_RAG_Evaluation.ipynb
Milestone 4: Fine-Tuning & V2	Tasks 4.1–4.5: LoRA Fine-Tuning, Hybrid RAG, Final Report	Fine_Tuning_Pipeline.ipynb, Solution_V2_...ipynb, Comparative_Analysis_Report.pdf